Quantum Optics Applications

(10 marks)

- Use the polarisation encoding

$$|0\rangle \equiv |H\rangle\,, \qquad |1\rangle \equiv |V\rangle\,. \tag{1}$$

The Pauli operators are defined by

$$X\,|0\rangle = |1\rangle\,, \qquad X\,|1\rangle = |0\rangle\,, \tag{2}$$

$$Z\,|0\rangle = |0\rangle\,, \qquad Z\,|1\rangle = -\,|1\rangle\,. \tag{3}$$

The four Bell states are

$$|\Phi^{\pm}\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \tag{4}$$

$$|\Psi^{\pm}\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}. \tag{5}$$

Question 1: Photonic qubits and Bell states

(2 marks)
Consider a pair of polarisation-entangled photons prepared in the state

$$|\Phi^{+}\rangle_{AB} = \frac{|H\rangle_A\,|H\rangle_B + |V\rangle_A\,|V\rangle_B}{\sqrt{2}}. \tag{6}$$

(a) Bell-basis orthogonality

Show that the four Bell states are mutually orthonormal.

(b) Local state of one photon

Construct the density operator

$$\rho_{AB} = |\Phi^{+}\rangle\langle\Phi^{+}| \tag{7}$$

and calculate the reduced state

$$\rho_A = \mathrm{Tr}_B\left(\rho_{AB}\right). \tag{8}$$

Explain what this reduced state implies about the outcome of a polarisation measurement performed on photon A alone.

(c) Correlations in two bases

Define the diagonal-polarisation states

$$|D\rangle = \frac{|H\rangle + |V\rangle}{\sqrt{2}}, \tag{9}$$

$$|A\rangle = \frac{|H\rangle - |V\rangle}{\sqrt{2}}. \tag{10}$$

Rewrite $|\Phi^{+}\rangle$ in the basis

$$\{|D\rangle\,, |A\rangle\}\,. \tag{11}$$

Describe the correlations observed when both photons are measured:

(i) in the $\{|H\rangle\,, |V\rangle\}$ basis; and

(ii) in the $\{|D\rangle\,, |A\rangle\}$ basis.

Question 2: Superdense coding with polarisation qubits

(4 marks)
Alice and Bob initially share the Bell state

$$|\Phi^+\rangle_{AB}\,. \tag{12}$$

Alice wishes to send one of the classical messages

$$00, \qquad 01, \qquad 10, \qquad 11. \tag{13}$$

She applies one of the operations

$$I, \qquad X, \qquad Z, \qquad XZ \tag{14}$$

to her photon and then sends that photon to Bob.

(a) Encoding the messages

For each of the four operations, calculate the resulting two-photon state:

$$(I \otimes I)\,|\Phi^+\rangle\,, \tag{15}$$
$$(X \otimes I)\,|\Phi^+\rangle\,, \tag{16}$$
$$(Z \otimes I)\,|\Phi^+\rangle\,, \tag{17}$$
$$(XZ \otimes I)\,|\Phi^+\rangle\,. \tag{18}$$

Hence complete a table containing:

- the classical message;
- Alice's operation; and
- the resulting Bell state.

Choose a consistent assignment of the four classical messages to the four operations.

(b) Information transmitted

Explain why an ideal Bell-state measurement allows Bob to recover two classical bits even though Alice sends only one photon.

(c) Linear-optical Bell-state measurement

Bob attempts to identify the Bell state by sending the two photons into the two input ports a and b of a balanced beam splitter.
For each polarisation

$$\lambda \in \{H, V\}, \tag{19}$$

use the beam-splitter transformations

$$\hat{a}^\dagger_\lambda \longrightarrow \frac{\hat{c}^\dagger_\lambda + \hat{d}^\dagger_\lambda}{\sqrt{2}}, \tag{20}$$
$$\hat{b}^\dagger_\lambda \longrightarrow \frac{\hat{c}^\dagger_\lambda - \hat{d}^\dagger_\lambda}{\sqrt{2}}. \tag{21}$$

Determine the output photon-number and polarisation patterns for the four input states

$$|\Phi^+\rangle\,, \qquad |\Phi^-\rangle\,, \qquad |\Psi^+\rangle\,, \qquad |\Psi^-\rangle\,. \tag{22}$$

Assuming that Bob has polarising beam splitters and photon-number-resolving detectors at the outputs, identify:

(i) which Bell states can be distinguished unambiguously; and

(ii) which Bell states remain indistinguishable.

(d) Decoding efficiency

For four equally likely messages, what is the probability that the standard passive linear-optical measurement identifies Alice's message unambiguously?
Briefly explain why this prevents deterministic dense coding using this measurement alone.

Question 3: BB84 quantum key distribution

(4 marks)
Alice implements the BB84 protocol using single photons. For each photon, she randomly chooses a bit value and one of two polarisation bases:

$$Z = \{|H\rangle\,, |V\rangle\}\,, \tag{23}$$
$$X = \{|D\rangle\,, |A\rangle\}\,, \tag{24}$$

where

$$|D\rangle = \frac{|H\rangle + |V\rangle}{\sqrt{2}}, \tag{25}$$
$$|A\rangle = \frac{|H\rangle - |V\rangle}{\sqrt{2}}. \tag{26}$$

Bob independently and randomly chooses the Z or X basis for each measurement.

(a) State preparation

Construct a table showing the quantum state Alice sends for each combination of bit value and basis.

(b) Basis sifting and loss

After the transmission, Alice and Bob publicly announce which bases they used. They retain only the results for which they used the same basis. These retained bits form the *sifted key*.
What fraction of Bob's detections are expected to be retained?
If Alice sends N photons through a channel with transmission η_{ch}, and Bob's detector has efficiency η_d, calculate:

(i) the expected number of photons detected by Bob; and

(ii) the expected length of the sifted key.

Neglect dark counts and all other imperfections.

(c) Intercept–resend attack

Suppose Eve intercepts every photon, measures it in a randomly chosen Z or X basis, and sends Bob a new photon prepared according to her measurement outcome.
The *quantum bit error rate* (QBER) is the fraction of sifted-key bits for which Bob's bit differs from Alice's bit:

$$Q = \frac{N_{\text{error}}}{N_{\text{sift}}}. \tag{27}$$

Because the QBER is calculated using the sifted key, Alice and Bob are already known to have used the same basis. Calculate the QBER caused by Eve using

$$\begin{aligned} Q = {} & P(\text{Eve uses Alice's basis})P(\text{error} \mid \text{Eve uses Alice's basis}) \\ & + P(\text{Eve uses a different basis})P(\text{error} \mid \text{Eve uses a different basis}). \end{aligned} \tag{28}$$

Determine each probability appearing in this expression.

(d) Detecting the attack and completing the protocol

Alice and Bob estimate the QBER by publicly comparing m randomly selected bits from their sifted key.
Assuming that errors in different bits are independent, the probability that they observe no errors is

$$P(\text{no errors}) = (1 - Q)^m. \tag{29}$$

Therefore, the probability that they detect at least one error is

$$P(\text{at least one error}) = 1 - (1 - Q)^m. \tag{30}$$

(i) Using the QBER calculated in part (c), calculate the probability that Alice and Bob observe at least one error when they compare $m = 20$ sifted-key bits.

(ii) Explain why the bits publicly revealed during this test cannot be included in the secret key.

(iii) After estimating the QBER, Alice and Bob perform error correction and privacy amplification. Briefly state the purpose of each step.